

=== CONFIGURACION GNB (1780307521) ===

```
# gNB configuration for srsRAN Project
#
# This configuration is for a 5G SA gNB.
#
# For more details on the configuration options, please refer to the srsRAN Project documentation.

# General gNB settings
gnb:
  mcc: 001
  mnc: 01
  nci: 0x000000001
  id: 0x1
  tac: 7
  # List of AMF addresses
  amf_addr: 10.53.1.2
  # gNB bind address for N2 interface
  bind_addr: 10.53.1.3

# Radio Frequency (RF) settings
rf:
  device_name: uhd
  device_args: ""
  # TX/RX gain for the RF device
  tx_gain: 70
  rx_gain: 40
  # Sample rate for the RF device
  srate: 38.4e6
  # Number of antennas
  nof_antennas: 1

# NR Cell configuration
cell_cfg:
  # NR band
  bands:
    - 78
  # Downlink NR ARFCN
  dl_nr_arfcn: 630000
  # SSB NR ARFCN (can be same as dl_nr_arfcn or slightly offset)
  ssb_nr_arfcn: 630000
  # Subcarrier spacing for common channels (SSB)
  ssb_scs: 30
  # Subcarrier spacing for data channels
  common_scs: 30
  # Number of physical resource blocks (PRBs)
  nof_prb: 106 # Corresponds to 20 MHz bandwidth with 30 kHz SCS
  # Cell radius in meters (for pathloss model)
  cell_radius: 100

# ZMQ interface for connecting to srsUE
zmq:
  # ZMQ bind address for gNB
  bind_addr: 0.0.0.0
  # ZMQ TX port (for sending to UE)
  tx_port: 2000
  # ZMQ RX port (for receiving from UE)
  rx_port: 2001
  # ZMQ RX address (for receiving from UE)
  rx_addr: 0.0.0.0

# Logging settings
log:
  all_level: info
  filename: /tmp/gnb.log
  file_max_size: -1
```

=== CONFIGURACION UE (1780307521) ===

```
# UE configuration for srsRAN Project
#
# This configuration is for a 5G SA UE.
#
# For more details on the configuration options, please refer to the srsRAN Project documentation.
```

```
[rf]
# Frequency offset for RF device (usually 0)
freq_offset = 0
# TX/RX gain for the RF device
tx_gain = 10
rx_gain = 5
# Sample rate for the RF device
srate = 38.4e6
# Number of antennas
nof_antennas = 1
# RF device name
device_name = uhd
# RF device arguments
device_args = sync=internal
# Time advance samples (auto for automatic calculation)
time_adv_nsamples = auto
```

```
[rat.nr]
# NR bands supported by the UE
bands = 78
# Number of carriers
nof_carriers = 1
# Maximum number of physical resource blocks (PRBs)
max_nof_prb = 106
# Number of physical resource blocks (PRBs)
nof_prb = 106
# Downlink NR ARFCN
dl_nr_arfcn = 630000
# SSB NR ARFCN
ssb_nr_arfcn = 630000
```

```
[pcap]
# Enable PCAP logging (none, mac, nas, all)
enable = none
# Filename for MAC layer PCAP
mac_filename = /tmp/ue_mac.pcap
# Filename for NR MAC layer PCAP
mac_nr_filename = /tmp/ue_mac_nr.pcap
# Filename for NAS layer PCAP
nas_filename = /tmp/ue_nas.pcap
```

```
[log]
# Global log level
all_level = info
# PHY library log level
phy_lib_level = none
# Hex dump limit for logging
all_hex_limit = 32
# Log filename
filename = /tmp/ue.log
# Maximum log file size (-1 for unlimited)
file_max_size = -1
```

```
[usim]
# USIM mode (soft for software-based USIM)
mode = soft
# USIM algorithm
algo = milenage
```

```
# Operator Code (OPC)
opc = 63BFA50EE6523365FF14C1F45F88737D
# K value (secret key)
k = 00112233445566778899aabbccddeeff
# International Mobile Subscriber Identity (IMSI)
imsi = 001010000000001
# International Mobile Equipment Identity (IMEI)
imei = 353490069873319
```

```
[rrc]
# 3GPP Release version
release = 15
# UE Category
ue_category = 4
```

```
[nas]
# Access Point Name (APN)
apn = srsapn
# APN protocol (ipv4, ipv6, ipv4v6)
apn_protocol = ipv4
```

```
[gw]
# Gateway configuration (no ip_addr as per rules)
```

```
[gui]
# Enable GUI
enable = true
```

```
[zmq]
# ZMQ bind address for UE
bind_addr = 0.0.0.0
# ZMQ TX port (for sending to gNB)
tx_port = 2001
# ZMQ RX port (for receiving from gNB)
rx_port = 2000
# ZMQ RX address (for receiving from gNB)
rx_addr = 0.0.0.0
```

=== DOCKER COMPOSE (1780307521) ===

```
# Copyright 2021-2026 Software Radio Systems Limited
#
# This file is part of srsRAN Project - https://github.com/srsran/srsran
#
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```

version: '3.8'

services:

```
open5gs_core:
  container_name: open5gs_core
  build:
    context: open5gs
    target: open5gs
  privileged: true
  networks:
    ran:
      ipv4_address: 10.53.1.2
  volumes:
    - ./open5gs/config:/open5gs/config
  command: /bin/bash -c "open5gs-amf & open5gs-ausf & open5gs-nrf & open5gs-pcf & open5gs-smf & open5gs-udm & open5gs-ued"
  environment:
    TZ: Europe/Madrid
```

```
srsran_gnb:
  container_name: srsran_gnb
  build:
    context: srsran
    target: srsran
  privileged: true
  networks:
    ran:
      ipv4_address: 10.53.1.3
  volumes:
    - ./srsran/config:/etc/srsran
  command: srsgnb -c /etc/srsran/gnb.yaml
  environment:
    TZ: Europe/Madrid
```

networks:

```
ran:
  driver: bridge
  ipam:
    config:
      - subnet: 10.53.1.0/24
```